# The University of Texas at Austin
# In-Class Work 18

*M378K Introduction to Mathematical Statistics*

November 14, 2026

## Consistency.

**Definition 18.1**: *$\hat{\theta}_n$ is said to be a* consistent estimator *of $\theta$ if*

$$\hat{\theta}_n \to \theta \quad \text{in probability as} \quad n \to \infty \ ,$$

*i.e., if for any $\varepsilon > 0$,*

$$\lim_{n\to\infty} \mathbb{P}\left[\left|\hat{\theta}_n - \theta\right| > \varepsilon\right] = 0.$$

**Theorem 18.2**: *Let $\hat{\theta}_n$ be* unbiased *and such that*

$$\text{Var}\left[\hat{\theta}_n\right] \xrightarrow{n\to\infty} 0.$$

*Then, $\hat{\theta}_n$ is a* ***consistent*** *estimator.*

**Problem 18.3**: *Let $Y_1, Y_2, ..., Y_n$ be a random sample from any distribution with finite first and second moments. Propose a consistent estimator for the population mean $\mu$ and* ***prove*** *that it is, indeed, consistent.*

**Solution**: *A natural choice is the sample mean*

$$\bar{Y}_n = \frac{1}{n}(Y_1 + Y_2 + ... + Y_n).$$

*We have already verified earlier in class that $\bar{Y}$ is* unbiased *for $\mu$. According to the theorem above, it suffices to show that*

$$\text{Var}\left[\bar{Y}_n\right] \xrightarrow{n\to\infty} 0.$$

*However, we know that*

$$\text{Var}\left[\bar{Y}_n\right] = \frac{\sigma^2}{n} \xrightarrow{n\to\infty} 0$$

*which completes our proof.*

**Problem 18.4**: *Consider a random sample $Y_1, Y_2, ..., Y_n$ from a power distribution with the density of the form*

$$f_Y(y) = \theta y^{\theta-1} 1_{(0,1)}(y).$$

*What is a consistent estimator for $\frac{\theta}{\theta+1}$? **Prove** that your choice is indeed consistent.*

**Solution**: *The expression $\frac{\theta}{\theta+1}$ stands for the population mean. Indeed,*

$$\mathbb{E}[Y_1] = \int_0^1 y f_Y(y)\,\mathrm{d}y = \int_0^1 \theta y^{\theta}\,\mathrm{d}y = \frac{\theta}{\theta+1}\,.$$

*So, a sensible choice for a consistent estimator is $\bar{Y}_n$. It's evidently unbiased. Let's try to use the previous criterion in terms of the vanishing variance. We know that*

$$\mathrm{Var}\,\left[\bar{Y}_n\right] = \frac{\mathrm{Var}\,[Y_1]}{n}\,.$$

*So, it suffices to verify that the variance of $Y_1$ is finite. We have*

$$\mathrm{Var}\,[Y_1] = \mathbb{E}[Y_1^2] - \left(\frac{\theta}{\theta+1}\right)^2.$$

*On the other hand,*

$$\mathbb{E}[Y_1^2] = \int_0^1 y^2\left(\theta y^{\theta-1}\right)\mathrm{d}y = \theta\int_0^1 y^{\theta+1}\,\mathrm{d}y = \frac{\theta}{\theta+2}\,.$$

*Clearly, $\mathrm{Var}\,[Y_1]$ is finite and the theorem applies. Hence, $\bar{Y}_n$ is, indeed, consistent.*